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Review

Hospital Length-of-Stay Prediction Using Machine Learning Algorithms—A Literature Review

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Abstract: Predicting hospital length of stay is critical for efficient hospital management, enabling proactive resource allocation, the optimization of bed availability, and optimal patient care. This paper explores the potential of machine learning algorithms to revolutionize hospital length-of-stay predictions, contributing to healthcare efficiency and patient care. The main objective is to identify the most effective machine learning algorithm for building a predictive model capable of predicting hospital length of stay. The bibliographic search of the existing literature on machine learning algorithms applied to hospital length of stay predictions highlighted the most relevant papers within this area of research. The papers were analyzed in terms of model types and metrics that contributed to the considerable impact on healthcare decision making. We also discuss the challenges and limitations of machine learning algorithms for predicting length of stay, and the importance of data quality and ethical considerations.

Keywords: hospital length of stay; prediction models; machine learning algorithms

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Citation: Almeida, G.; Brito Correia, F.; Borges, A.R.; Bernardino, J. Hospital Length-of-Stay Prediction Using Machine Learning Algorithms—A Literature Review. *Appl. Sci.* **2024**, *14*, 10523. https://doi.org/10.3390/app142210523

Academic Editor: Maria Rizzi

Received: 25 September 2024
Revised: 7 November 2024
Accepted: 11 November 2024
Published: 15 November 2024

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1. Introduction

This paper offers a comprehensive overview of machine learning (ML) algorithms and their applications in the healthcare industry, with an emphasis on hospital length of stay (LoS) prediction. The main objective of this study is to analyze which ML algorithms are the most effective to build a predictive model capable of predicting hospital LoS.

Initially, different types of ML algorithms are explored, detailing their characteristics and the importance of the most relevant features used for predicting hospital admissions and hospital LoSs. Afterwards, we analyze how these algorithms are used to improve outcomes and efficiency in healthcare. Ethical considerations, such as patient privacy, data management frameworks, and bias mitigation are important for these types of studies, so they are also addressed. These topics provide an overview of the complexities and responsibilities involved in integrating ML into healthcare, ensuring that technologies are not only effective, but also equitable and respectful of patients' rights.

Hospital admissions and hospital LoSs are multifaceted challenges that involve a complex interplay of patient health, resource availability, and healthcare policies. Accurately predicting admissions or LoSs is notoriously difficult, yet it holds immense potential for improving healthcare system efficiency, reducing costs, and ultimately improving patient outcomes. This article delves into the intricate world of hospital LoS prediction with the goal of providing readers with a thorough understanding of the subject. Among the myriad issues addressed are inaccurate predictions, data complexity, resource constraints, patient variability, and ethical considerations.

The contributions offered by this review paper include improved prediction accuracy, enhanced decision support, cost reduction initiatives, and the development of an ethical framework.

This article focuses on the use of ML algorithms to predict hospital LoS. Therefore, four research questions (RQ) were formulated:

- RQ1: What are the key factors that significantly influence hospital LoS?
- RQ2: What types of ML algorithms have been employed for hospital LoS predictions?
- RQ3: How do these algorithms compare in terms of performance metrics to predict hospital LoSs?
- RQ4: What are the potential benefits and limitations of using ML for hospital LoS predictions in healthcare settings?

This paper is a review on the prediction of hospital LoS using ML algorithms. Section 2 starts with a brief overview of ML algorithms and their use in healthcare systems, followed by their use for hospital LoS predictions. Section 3 describes the ethical considerations we must pay attention to when using these types of data, and also when using ML algorithms. Section 4 describes the Kitchenham's methodology used and the work developed using it, up to an analysis of twelve selected research articles relevant to the topic using this methodology. It includes a detailed outline of the research protocol, covering the data source, search string, prerequisites, and inclusion and exclusion criteria. Section 5 presents the results provided by each study. Section 6 provides an analysis and discussion of the results. Finally, Section 7 concludes the article by summarizing and highlighting the potential of leveraging ML applied to hospital LoS prediction, ultimately promoting improved healthcare delivery. Through this review, the paper aims to provide readers with a comprehensive understanding of the potential advantages and challenges associated with hospital LoS predictions using ML algorithms.

2. Machine Learning Algorithms in Healthcare

ML algorithms have become pivotal tools in various fields [1], including healthcare [2,3]. Also, they can be broadly categorized into several types, including supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning [4]. Examples of supervised learning algorithms are linear regression, random forest, and support vector machine [5]. These algorithms are among the most used for healthcare modeling development, as specified in this article [6], which discusses their effectiveness in various healthcare applications.

ML algorithms are a class of algorithms that learn from data and can be used for a variety of tasks, including predicting hospital admissions and predicting hospital LoSs. The application of ML algorithms can be divided into the following steps [7]:

- Data gathering [8]: The data needed to train ML algorithms can be collected using various sources, namely hospital information systems, electronic medical records systems, and patient surveys;
- Data preparation: The previously gathered data must be prepared for training the algorithms, and this includes cleaning, removing null values, and standardizing the data [9];
- Choice and consideration of algorithms to use: There are many ML algorithms, so it is necessary to choose the most appropriate one in terms of the prediction to be made;
- Algorithm training: ML algorithms are trained and use previously prepared data. Training involves presenting data to the algorithms and evaluating their prediction ability;
- Improve the model: Constantly try to improve the model and obtain better results based on the data provided.

2.1. Application of Machine Learning Algorithms in Healthcare

Artificial intelligence can be used for so much these days, and one area that can benefit is healthcare [10]. One notable application of ML in healthcare is for the early detection and treatment of sepsis, a life-threatening condition. In a randomized clinical trial, a machine

learning-based algorithm predicted severe sepsis with greater accuracy than traditional methods, such as SOFA and SIRS scoring systems. This resulted in significant reductions in both hospital length of stay (20.6%) and in-hospital mortality (58%), highlighting the transformative potential of ML to improve patient outcomes through early intervention [11].

Some of the main applications of ML algorithms in healthcare include:

- Diagnosis: ML algorithms can be used to analyze data from medical exams, such as images, laboratory tests, and clinical data, with the aim of identifying patterns that may indicate the presence of a disease, for example;
- Personalized medicine: The patient can be analyzed and be administered personalized medicine based on his condition;
- Treatment: ML algorithms can also be used to personalize the treatment of patients based on their individual characteristics and the history of their disease, which will consequently help to improve the effectiveness and safety of the treatment;
- Outbreak prediction: AI and machine learning are used to monitor and predict epidemics by analyzing vast data from satellites, social media, and websites, aiding in disease outbreak predictions, especially in countries in the Global South.

Despite the revolutionary potential for healthcare, the application of ML algorithms also faces some challenges, such as:

- Bias is present in ML;
- Lack of quality data;
- Data annotation must be performed accurately;
- Need for hyperparameter tuning.

One of the main challenges is the need for large datasets to train the algorithms. Also, algorithms that are developed should be validated in an ethical and responsible way. The opportunities offered by ML are significant, where the technology has the potential to improve the quality of medical care, reduce costs, and increase the efficiency of the healthcare system, which decreases the risk relative to the reward.

Especially when dealing with complex multivariate time-series data, ML algorithms have become increasingly important for predicting hospital LoS, due to algorithms such as random forest and support vector machine. This is due in part to their robustness and processability with high-dimensional data applications in healthcare, where these models are used to predict patient outcomes using various clinical criteria of variables, such as patient demographics and vital signs. Random forest is particularly effective because of its ability to capture complex interactions that require intensive preprocessing due to its invariance. Support vector machine, on the other hand, excels at classification tasks with smaller datasets and can handle outliers more efficiently. A comparative study of these algorithms, as performed by Ruiz et al. [12], shows versatility and efficiency in a variety of industries, in terms of time-series classification. Their results show that random forest, support vector machine, and other algorithms are still very competitive when applied to time-series data, making them suitable for clinical LoS prediction tasks, where time series and patient changes must be accurately modeled.

2.2. Application in Predicting Hospital Length of Stay

ML algorithms are used by almost 60% of companies in the healthcare industry [13]. These algorithms play a critical role in predicting hospital LoS with the most-used types being classification algorithms, regression algorithms, and neural network algorithms.

- Classification algorithms can be employed to predict whether a patient will be admitted to the hospital (binary classification). These algorithms analyze various patient data to determine the likelihood of admission or the hospital LoS. Popular classification algorithms include decision trees, random forest, and logistic regression. For example, logistic regression can effectively handle binary outcomes, making it suitable for predicting hospital admissions based on patient health indicators and history;

- Regression algorithms can be used to forecast the number of days a patient will remain hospitalized: the hospital LoS. Unlike classification, regression aims to predict a continuous numeric output. Common regression algorithms include logistic regression, ElasticNet regression, and support vector regression. These models consider a multitude of variables, such as patient demographics, clinical conditions, and treatment plans, to provide accurate hospital LoS predictions. Such predictions are crucial for optimizing bed management and resource allocation in hospitals.

3. Ethical Considerations

As ML technologies continue to advance, their application in healthcare and hospitals presents profound opportunities for improving patient outcomes and operational efficiencies. However, the deployment of these technologies also brings forth a range of ethical considerations that must be carefully examined.

This section explores the key ethical issues associated with using data for ML prediction in healthcare settings, addressing the concerns related to patient privacy, data management frameworks, fairness and bias mitigation, informed consent, and data usage. Exploring these dimensions ensures that the integration of ML solutions is guided by principles that prioritize patient welfare and uphold the highest standards of ethical practice [14].

One significant ethical concern is fairness and bias mitigation, as ML algorithms can inadvertently propagate existing biases present in training data, leading to unequal treatment across different patient demographics [15]. Exploring these dimensions ensures that the integration of ML solutions is guided by principles that prioritize patient welfare and uphold the highest standards of ethical practice, and, by addressing these ethical considerations, healthcare providers can harness the power of ML technologies while fostering trust and accountability in their implementation.

3.1. Patient Privacy

As ML technologies continue to permeate healthcare, safeguarding patient privacy has emerged as a paramount concern. The integration of artificial intelligence in clinical settings offers substantial opportunities for enhancing patient outcomes, but it also raises significant ethical, legal, and technological challenges related to the management of sensitive patient data. A review by Williamson and Prybutok [16] emphasizes the need for robust frameworks that ensure patient privacy while enabling data-driven insights.

When leveraging ML algorithms to predict hospital LoS, patient privacy is a vital concern. These predictive models can use various types of sensitive patient data, including medical records, demographic information, and lifestyle factors, which raises several important privacy issues.

- Data security and unauthorized access: "94% of hospitals had at least one security breach in the past two years" were mentioned in [17]. Increased risk of data breaches and unauthorized access to sensitive patient information due to complex big data systems;
- Privacy risks of data aggregation: Combining data from multiple sources can lead to the re-identification of individuals;
- Challenges of data anonymization: Anonymization techniques can be flawed, risking patient re-identification;
- Data governance and compliance issues: Ensuring data management practices meet legal standards is challenging [18];
- Informed consent and ethical considerations: Patients must be clearly informed and provide consent for their data to be used;
- Data integrity and accuracy: Accurate and complete data are essential for both privacy and effective predictions [19].

3.2. Data Management Frameworks

Various data management frameworks guide hospitals in protecting patient privacy during predictive analytics, offering structured approaches to data management throughout its lifecycle. These frameworks ensure that patient data are treated safely and ethically. Some key frameworks include:

- Health Importability and Accountability Act: This organization is a comprehensive United States federal law designed to protect the privacy and security of patient health information. It sets standards for data protection that hospitals must follow to ensure patient information is handled confidentially and securely [20];
- General Data Protection Regulation: This organization is a European Union regulation that imposes strict guidelines on the collection, use, and storage of personal data, including patient data. Hospitals operating in the European Union or processing data from EU citizens are required to comply with "General Data Protection Regulation" to protect individual privacy and data security [21].

They offer essential guidelines and standards that support hospitals in managing patient data responsibly and safeguarding patient privacy in the context of predictive analytics.

3.3. Fairness and Bias Mitigation

Ensuring fairness and reducing bias in ML models is essential for equitable healthcare trained models for them to perform well [22]:

- Identifying bias. Detect biases in ML models and data:
  - Can be performed by utilizing some techniques, for example, bias audits (utilize metrics like the disparate impact ratio and equal opportunity or data analysis);
- Mitigating bias. Reduce identified biases in ML models:
  - Can be performed by the utilization of oversampling to balance data representation;
- Evaluating Impact. Assess the effects of biases on patient outcomes:
  - Can be performed by utilizing impact analysis that measures fairness impacts on different patient groups.

By addressing bias and ensuring fairness, we can improve the ethical application of ML technologies in healthcare.

The quality of data plays a crucial role in ensuring fairness and mitigating bias in machine learning models. This article emphasizes that clean, complete, and well-processed data are essential to avoid introducing biases that could disproportionately affect certain patient groups. Data preprocessing, including handling missing values and normalizing features, ensures that models are trained on equitable datasets. Without such measures, biases in the raw data—such as overrepresentation of specific demographics—could skew predictions, ultimately affecting the model's fairness in clinical settings [23].

3.4. Features to Consider in Hospital Admission and Hospital LoS Predictions

In the realm of predicting hospital admissions and LoSs, identifying the most relevant features is crucial for developing effective predictive models. Features that can significantly impact hospital admission and hospital LOS predictions are [24]:

1. Patient demographics, as also used in [25]. Demographic information provides a foundation for understanding patient backgrounds and potential hospital LoS variations:
   a. Age—Older patients may have longer stays due to slower recovery times;
   b. Gender—Gender differences can influence disease progression and treatment outcomes;
2. Admission Details [25]. Information about the admission process can indicate the severity and type of care required:
   a. Admission type—Emergency vs. elective admissions often correlate with different hospital LoS outcomes;

   b. Admission source—Whether the patient was referred from another facility or admitted directly;
3. Clinical Parameters. Clinical parameters capture the medical status of patients, which directly affects the hospital LoS:
   a. Diagnosis—Primary and secondary diagnoses influence treatment complexity and duration;
   b. Comorbidities—Presence of additional health conditions affect hospital LOS through increased treatment needs;
   c. Severity of illness;
4. Treatment and Procedures. Treatments and procedures performed during a patient's stay influence the length of their hospital admission:
   a. Procedures performed—Complexity and number of procedures can extend hospital LoS;
   b. Medication Regimens—Duration and type of medication impact recovery time;
5. Patient Functional Status. A patient's functional status can influence their recovery trajectory and hospital LoS:
   a. Functional independence—Measures of a patient's ability to perform daily activities;
   b. Rehabilitation needs—Extent of rehabilitation services required post-discharge.

Selecting the most important features for predicting hospital LOS involves a multi-faceted approach that incorporates patient demographics, clinical parameters, admission details, and other relevant factors. Each feature plays a distinct role in shaping hospital LoS outcomes, and effective feature selection is crucial for developing accurate and actionable predictive models in healthcare settings.

4. Materials and Methods

This section describes in detail the methodology used for conducting this review. It begins with the presentation of the protocol followed by a literature review, where the essential steps to ensure the quality and relevance of the data collected are discussed. The protocol includes the definition of research questions, the identification of data sources, the formulation of search strings, the preconditions necessary for the inclusion of studies, the inclusion and exclusion criteria that guide the selection of papers, and the extraction of the most relevant papers. Next, we cover the most used metrics, providing a solid basis for evaluating the results and comparing different studies. It concludes with an analysis.

4.1. Review Paper Protocol

This review followed Kitchenham's methodology [26] that is extensively used in computer science research. Principles and guidelines for developing systematic reviews are outlined in this methodology. The following steps were applied: (a) research question definition; (b) data source selection; (c) search string definition; (d) precondition definition; (e) inclusion and exclusion criteria definitions; and (f) data extraction.

4.1.1. Data Source

In the Google Schoolar search engine, users can look for books, articles, reports, theses, and academic publications in a variety of subject areas. In this review, articles were searched for by using Google Scholar.

4.1.2. Search String

The search string used was (hospital AND prevision AND forecast AND "machine learning" AND predict AND "length of stay" AND LoS-"COVID-19"-malnutrition-transfusion-review). Four words/expression were removed from the search when re-searching the articles: "COVID-19", "malnutrition", "transfusion", and "review". The

reason for this was to narrow the search, since the intention was to predict hospital LoS in general. With this search string, 604 articles were found.

4.1.3. Preconditions

Establishing preconditions is essential for developing a methodical and organized strategy for article extraction. To guarantee the accuracy and applicability of the retrieved articles, the following prerequisites were carefully analyzed:

- Language: Studies must be written in English. Considering only articles in English, 558 articles remained.
- Publication Date. Studies must be published after 2020. Considering only articles, this last precondition left 315 articles.

4.1.4. Inclusion and Exclusion Criteria

To ensure that the selected studies aligned with the research goals and to address the specific context of ML for hospital LoS predictions, the following inclusion criteria were defined:

- Research focus: Studies must investigate the application of ML algorithms to predict hospital LoS;
- Publication type: Studies must be published in peer-reviewed journals/conferences.
- The exclusion criteria defined were:
- The article does not refer to ML algorithms;
- The article does not focus on hospital LoS predictions;
- The article is not available (for example, due to institution limitation access);
- Duplicate study;
- Documents other than articles are not considered (like theses and books).

4.1.5. Data Extraction

With all this in mind, were obtained n = 604 articles, n being the number articles. If we selected only articles written in English, n = 558 articles remained. Including both preconditions, the language and the publication date (after 2020) left n = 315 articles.

Afterward, some inclusion and exclusion criteria were considered, titles were analyzed, leaving n = 52 articles. The abstracts of these selected articles were then analyzed, and only n = 21 articles were left. Finally, 7 articles were excluded because they were not available, and the search ended with n = 12 articles to be analyzed in this review. Figure 1 illustrates the process used in this literature review.

4.2. Most Commonly Used ML Algorithms in Healthcare

In the healthcare field, ML algorithms have been widely used to address various challenges, especially forcing patient outcomes, such as hospital LoS, disease diagnosis, and treatment plan. This also depends on evaluating their performance through various metrics measuring accuracy, precision, recall, and usability.

The following Table 1 shows the most commonly used algorithms in healthcare.

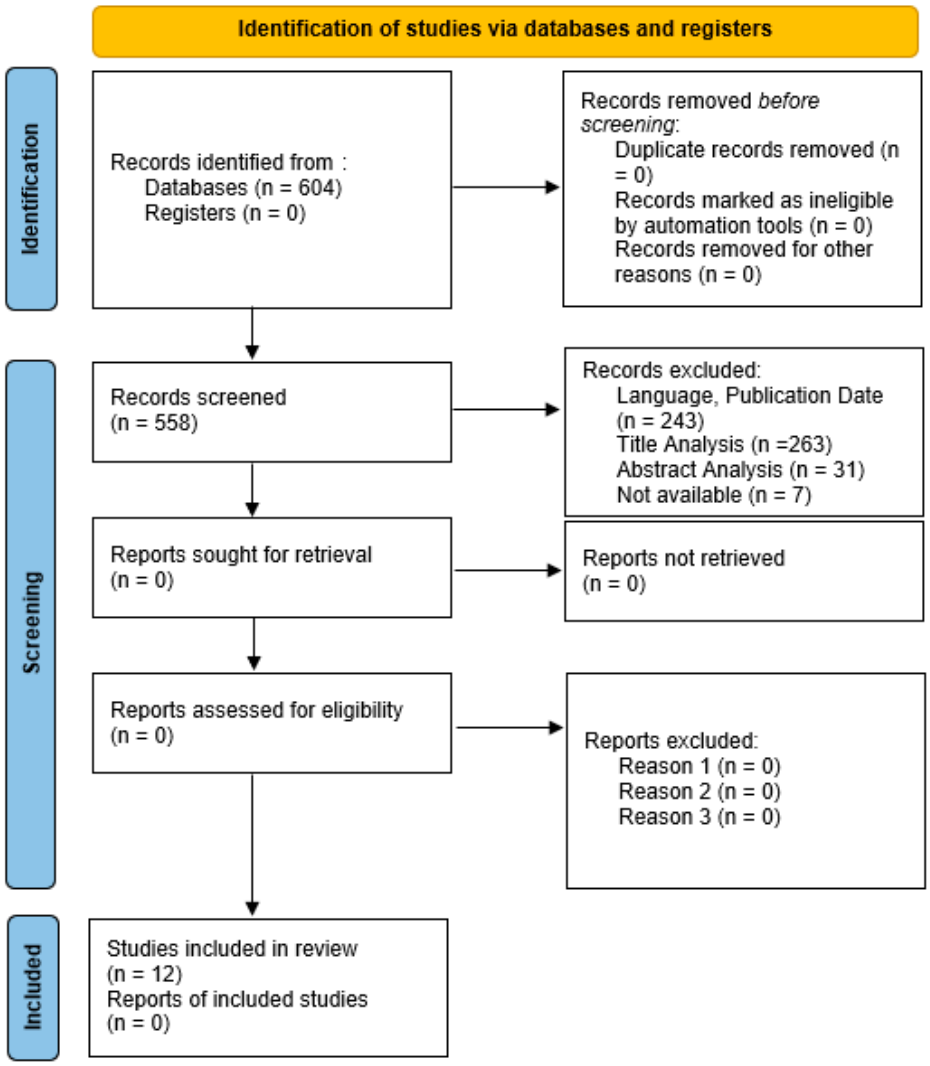

Figure 1. Literature review process.

Table 1. Most-used ML algorithms in healthcare.

ML Algorithm	Type of Algorithm	Frequency of Use	Common Applications
Linear Regression	Supervised (Regression)	High	Predictive modeling, trend analysis
Logistic Regression	Supervised (Classification)	High	Binary classification, medical diagnostics
Decision Trees	Supervised (Classification/Regression)	Moderate	Data mining, risk analysis, decision support systems
Random Forest	Supervised (Ensemble Learning)	Very High	Fraud detection, customer churn prediction, genomics
K-Nearest Neighbors	Supervised (Classification/Regression)	Moderate	Pattern recognition, recommendation systems
Support Vector Machines	Supervised (Classification)	Moderate to High	Image recognition, text classification, bioinformatics
Naive Bayes	Supervised (Classification)	Moderate	Spam filtering, sentiment analysis

4.3. Most Important Metrics for Hospital LoS Prediction

To develop effective hospital LoS prediction models, it is essential to evaluate their performance using various metrics that reflect different aspects of the model's accuracy and utility.

This subsection explores the most important metrics used for evaluating hospital LoS prediction models. Each metric serves to measure different dimensions of the model's performance, ranging from predictive accuracy to the practical impact on healthcare operations. Understanding these metrics is fundamental for selecting the right model for specific healthcare scenarios and for guiding ongoing improvements to predictive strategies.

- Mean Absolute Error (*MAE*)—measures the average magnitude of the errors in predictions, without considering their direction. It is the average over the test sample of the absolute differences between the prediction and actual observation where all individual differences have an equal weight;

  The *MAE* is obtained using the following formula:

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|\hat{y} - y_1|, \tag{1}$$

- Accuracy—a fundamental metric used to evaluate the performance of classification models, including those used for predicting hospital LoS in healthcare settings. It measures the proportion of correct predictions made by the model out of the total number of predictions. In the context of regression models, accuracy can be adapted to evaluate how often the predicted hospital LoS falls within a specific acceptable range of the actual hospital LoS;

  The accuracy of a classification model is defined as:

$$Accuracy = (TP + TN)/(TP + TN + FP + FN), \tag{2}$$

- R^2 Score—a statistical metric used to evaluate the goodness of fit of a regression model. It measures the proportion of the variance in the dependent variable (e.g., hospital LoS) that is predictable from the independent variables. In the context of hospital LoS prediction, R^2 indicates how well the model's predictions approximate the actual outcomes [27].

The coefficient of determination, R^2, is a measure of how well the model explains the variability of the dependent variable. It is calculated by the following formula:

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(\hat{y} - \overline{y})^2}{\sum_{i=1}^{N}(y_i - \overline{y})^2}, \tag{3}$$

- *F1 Score*—a metric used to evaluate the performance of classification models, especially when dealing with imbalanced classes. It is the harmonic mean of precision and recall, providing a single score that balances both the concerns of false positives and false negatives.

  The formula for the F1 score is:

$$F1\ Score = 2 \times \frac{Precision * Recall}{Precision + Recall}, \tag{4}$$

- Root Mean Squared Error (*RMSE*)—a widely used metric for evaluating the performance of regression models. It measures the average magnitude of the errors between predicted and actual values, taking the square root of the average squared differences. The *RMSE* provides a way to quantify how far off a model's predictions are from the actual outcomes, and it is useful for comparing different models or understanding model performance in predicting hospital LoS.

The formula for the *RMSE* is:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{y} - y_i)^2}, \quad (5)$$

Understanding these metrics is essential for selecting the most appropriate model for specific healthcare scenarios and guiding the refinement of predictive strategies. Each metric offers unique insights into the model's performance, helping developers to evaluate not only the accuracy of predictions, but also their practical implications in healthcare settings.

4.4. Analysis of Selected Papers

We analyzed 12 articles. These articles focused on ML applied to hospital LoS prediction.

ML is a subset of predictive business analytics that can empower healthcare providers to analyze large datasets, helping them to uncover valuable insights useful for the process of making decisions. The 12 articles that were analyzed here were obtain using Kitchenham's methodology [26], described in Section 3 in this work.

In the study carried out in [28], the authors aim to develop a decision support system to assist physicians in predicting hospital LoS using various regression models (Table 2). Utilizing the MIMIC-II dataset (Table 1), the authors compared the performance of several algorithms (Table 2). Their methodology included data preprocessing, such as converting categorical features to numerical values, and normalizing the data. The study found that regularization techniques, particularly ElasticNet, outperform simple linear regression in terms of the mean absolute error (MAE), highlighting the importance of regularization for preventing overfitting and improving model generalizability.

The study's strength lies in its practical application, evident motivation, and comprehensive comparative analysis of different regression models. Their research demonstrates practical relevance and usefulness by utilizing real-world datasets.

This study makes a significant contribution to the field of healthcare management and predictive analytics by demonstrating the potential of regularization techniques in enhancing prediction accuracy.

Article [29] develops a system to predict hospital LoS using neural networks, aiming to enhance efficient hospital resource management in Indonesia. Using a dataset from UPT Puskesmas Arjasa Kangean, the authors analyzed 3055 patient records with 30 attributes, implementing a neural network for prediction. They emphasized the importance of data preprocessing, including nominal to binary transformations, using one-hot encoding, and feature selection with the random forest algorithm. The optimal model configuration, determined through grid search and random search techniques, achieved an accuracy of 94.74% with the following specific parameters: Epoch: 50, hidden units: 52, batch size: 4000, Adam optimizer, and linear activation.

The application of the neural network in this study enhances hospital management by predicting hospital LoS accurately, thus aiding in better resource allocation and operational efficiency. However, this study could be improved by expanding on the interpretability of the neural network model, such as analyzing the significance of different features in the prediction. Additionally, testing the model on diverse datasets from other hospitals would enhance its generalizability. Overall, the research provides a valuable contribution to healthcare informatics by showcasing the practical utility of neural networks when predicting patient hospital LoS and improving hospital efficiency through advanced data analytics and visualization techniques.

In article [30], the authors explore the use of ML models to predict the hospital LoS for neonates in the Neonatal Intensive Care Unit [31]. Utilizing the MIMIC-III database, which includes data from 8024 Neonatal Intensive Care Unit stays, the study emphasizes the significance of efficient resource allocation in critical care settings. The authors preprocessed the data through feature engineering, including the transformation of categorical variables using one-hot encoding and the imputation of missing values with the Asexual Reproduction Optimization algorithm. The study employs various regression models, namely

random forest, XGBoost, k-nearest neighbors, logistic regression, and gradient boosting, ultimately achieving a high R2 score of 0.78 using patient diagnoses and demographic data collected within the first 24 h of admission.

Table 2. Dataset overview.

Study	Dataset	Size and Features
[28]	MIMIC—II	4927 patients and 24 features
[29]	UPT Puskesmas Arjasa Kangean	3055 patient records, 30 features
[30]	MIMIC-III	8024 patient stays, various features
[32]	Electronic Health Records	Not specified, various features
[33]	MIT MIMIC-III	Not available, various features
[34]	Unspecified	350,393 records, 46 features
[35]	Medical Institution Dataset	Not specified, various features
[36]	Electronic Health Records	Not specified, 10 features
[37]	Femur Fracture Data	547 (pre-DTAP), 562 (post-DTAP), Various features
[38]	Total Hip Arthroplasty	792 samples, 13 features
[39]	Patient Admission Data	Not specified, various features
[40]	Neurosurgical Treatment Cases	90,685 cases, 6 features

This research demonstrates the practical application of data mining techniques in healthcare, particularly in enhancing operational efficiency and patient management in Neonatal Intensive Care Units. By leveraging ML, this study provides a valuable tool for predicting hospital LoS, thereby potentially helping in resource allocation and potentially reducing healthcare costs. Overall, this research significantly contributes to the field of predictive analytics in healthcare, offering insights that could lead to better patient outcomes and a more efficient use of medical resources.

The study in [32] focuses on predicting LoSs in hospitals, which is crucial for optimizing staffing and bed management. Utilizing Electronic Health Records available at admission, including patient demographics, diagnoses, and physiological test results, the study focuses on developing a predictive model applicable to a broad patient spectrum. Despite challenges like data entry delays and absent non-medical factors in electronic health records, the authors designed a data cleaning and integration process to construct a comprehensive dataset for predictive modeling. Through survival analysis, including semi-parametric and parametric models, they achieved similar performances across models, with a concordance index of approximately 0.7.

The findings suggest that patient information in electronic health records is sufficient for hospital LoS predictions, showing potential for routine predictions without additional data inputs. Future steps involve real-time data entry facilitation, automation of coding processes, and incorporation of social factors influencing discharge times. Overall, this study underscores the feasibility of leveraging electronic health record data for hospital LoS predictions, offering insights into enhancing hospital operations and resource management.

Research paper [33] introduces an ML-based solution to tackle the issue of patient overcrowding in hospital emergency departments. By utilizing decision trees algorithm and training on datasets presenting symptoms, diseases, severity conditions, and admission statuses, the model aims to predict the likelihood of patient admissions to emergency department wards. Through two main phases—disease prediction and admission status prediction—the model helps forecast the number of patients likely to require an admission, enabling hospitals to make necessary resource arrangements in advance. Experimental results indicate an overall accuracy of over 80%, highlighting the effectiveness of the model in aiding resource allocation and mitigating patient overcrowding in emergency departments.

In conclusion, the proposed ML model offers a promising approach to enhance emergency department management and patient care. Future directions include the further refinement of the model with additional features, such as predicting diseases admissions,

and exploring alternative algorithms to enhance performance and address evolving healthcare challenges effectively.

In [34], the short research paper outlines an ML approach to predict patient consultation length in emergency departments, crucial for anticipating operational bottlenecks and improving service provision. With emergency departments facing challenges like increased patient volumes and resource shortages, accurately estimating consultation times becomes essential to mitigate overcrowding and waiting times. Leveraging factors, such as patient demographics, vital signs, arrival time, and triage severity levels, the study proposes two ML models, random forest and neural network, to forecast consultation lengths. By training on a dataset spanning nearly a decade with over 350,000 entries, the models demonstrate improved accuracy when incorporating additional features beyond triage levels, such as patient symptoms, age, doctor's ID, and registration details.

Although this research does not perform hospital LoS predictions and focuses on predicting consultation lengths, it is considered important because of the most-used ML algorithms in healthcare subjects and matters.

In [35], the authors showed a significant advancement in healthcare analytics, particularly in emergency department management. By harnessing the power of ML algorithms, like logistic regression, modified random forest, and gradient boosting, the study demonstrates the potential to accurately predict whether a patient will require an inpatient admission at the time of the triage. This predictive capability has far-reaching implications, enabling healthcare providers to optimize resource allocation, streamline patient flow, and ultimately enhance the quality of care delivered in emergency settings.

Moreover, the comparison of different ML algorithms provides valuable insights into their relative performance in predicting hospital admissions. The superior performance of the gradient boosting algorithm underscores its effectiveness in handling complex medical data and extracting meaningful patterns.

In [36], the critical task of predicting whether patients require inpatient or outpatient treatment is addressed, particularly focusing on hospital intensive patients. By leveraging various ML techniques, such as logistic regression, decision trees, random forest, and gradient boosting, and, notably, neural network, the research aims to identify the most accurate predictive model. The methodology includes three key stages: data collection, data preprocessing, and data modeling. Through the utilization of the Electronic Health Record Predicting data obtained from private hospitals in Indonesia, this study examines features such as laboratory test results, patient age, and gender to make informed predictions regarding hospitalization decisions.

Among the ML algorithms evaluated, the neural network emerges as the most effective, boasting an accuracy rate of 74.47%. This surpasses the performance of other algorithms, like logistic regression, decision trees, random forest, and gradient boosting. This study contributes valuable insights into the application of ML in healthcare decision making, paving the way for further advancements in predictive analytics and patient management strategies.

The study in [37] aims to predict the hospital LoS for patients undergoing femur fracture surgery, particularly focusing on the implementation of a new diagnostic–therapeutic-assistance pathway. By utilizing multiple linear regression models, the researchers analyzed patient data collected before and after the adoption of the diagnostic–therapeutic-assistance pathway. Their findings revealed promising outcomes, with the multiple linear regression models showing a predictive capability when estimating the hospital LoS based on patient demographics, comorbidities, and pre-operative hospital LoS. The pre-diagnostic–therapeutic-assistance pathway model exhibited a slightly better performance compared to the post-diagnostic–therapeutic-assistance pathway model, indicating the potential impact of the new pathway on hospital LoSs.

Incorporating the insights from this study, this article can enrich the discussion by providing a real-world example of predictive modeling in healthcare. By showcasing the effectiveness of multiple linear regression models in estimating hospital LoSs for femur

fracture patients, this article underscores the practical applications of ML in healthcare management. This study highlights the importance of considering not only algorithmic performance, but also the contextual factors and interventions that may affect predictive modeling outcomes in healthcare settings.

The study in [38] presents a solution to predict hospital LoS after total hip arthroplasty, a critical factor influencing patient recovery and healthcare costs. Traditional prediction methods lack accuracy and generalization, prompting the adoption of the XGBoost algorithm, an ML technique. By analyzing patient data and optimizing model parameters, this study demonstrates that XGBoost outperforms traditional methods, with an average prediction error of 2.03 days. Integrating these findings into this article for hospital LoS predictions in hip arthroplasty patients could highlight the potential of advanced algorithms to improve patient care and resource allocation in orthopedic surgery.

Article [39] presents an ML-based data-mining model aimed at analyzing and predicting patient admissions, particularly in emergency departments. The authors highlight the challenges of emergency department crowding and emphasize the need for innovative approaches to improve patient flow and prevent overcrowding. They discuss the various ML algorithms employed, including naive Bayes, random forest, and support vector machine, to predict patient admissions based on features such as age, gender, blood pressure, diabetes, and previous medical records. This study also compares the performance of these algorithms and identifies the support vector machine as the most accurate predictor among them.

In the Results Section, we concluded that the support vector machine model outperforms the naive Bayes and random forest classifier when predicting patient admissions accurately. They suggest that these ML models can significantly benefit hospitals by addressing the challenge of patient overcrowding in emergency departments and improving overall patient flow.

In article [39], although the values are specified, the author concludes that the support vector machine was the most accurate model.

Study [40] focuses on evaluating the accuracy of predicting hospital LoS following neurosurgery using the GPT-3 language model in comparison to predictions made by both doctors and patients. Building on previous research, the study employed unstructured textual data from electronic health records to train the neural network language model. The researchers then compared the hospital LoS predictions generated by the model with the expectations of doctors and patients to assess its efficacy.

The findings revealed that there was no significant disparity (p = 0.109) between hospital LoS predictions made by doctors, patients, and the GPT-3 language model. However, neurosurgeons exhibited a tendency to offer more accurate prognoses compared to both the model and patients. Despite this, this study concluded that modern neural network language models, like GPT-3, show promise for hospital LoS predictions for neurosurgery patients.

5. Results

This section provides a comparative analysis of the results obtained from various ML algorithms applied to predict hospital admission and hospital LoS outcomes, based on a review of 12 selected studies. The comparison focuses on the datasets used, methods applied, metrics evaluated, and the performance outcomes achieved.

5.1. Overview of Datasets

The studies analyzed utilized diverse datasets, with varying sizes and features, to predict hospital LoS. Table 2 summarizes the datasets employed in these studies.

5.2. Overview of Algorithms Used and Results from Each Article

Different ML algorithms were evaluated across the studies, including regression models, neural network, decision trees, and gradient boosting. Table 3 presents the key performance metrics reported for each algorithm.

Table 3. Comparation of algorithms and metrics.

Study	Method	Algorithm	Metric	Values
[28]	Regression	Logistic Regression	MAE	198,379,877,732,011.9
		Ridge Regression	MAE	0.82131
		Lasso Regression	MAE	0.96865
		ElasticNet	MAE	0.95121
[29]	Neural Network	Default Parameters	Accuracy	94.66%
		Grid Search Optimization	Accuracy	94.74%
		Random Search Optimization	Accuracy	94.74%
[30]	Regression	Random Forest	R^2 Score	0.7780
		XGBoost	R^2 Score	0.7608
		Gradient Boosting Algorithm	R^2 Score	0.7651
		Logistic Regression	R^2 Score	0.6466
		K-Nearest Neighbors	R^2 Score	0.7306
[32]	Survival Analysis	Various	Concordance Index	0.7
[33]	Decision Trees	Decision Trees	Accuracy	Above 80%
[34]	Regression	Random Forest	Accuracy	59.78%
				36.57%
	Neural Network	Neural Network	Accuracy	47.52%
				36.67%
[35]	Classification	Logistic Regression	Accuracy	80.54%
		Modified Random Forest	Accuracy	81.09%
		Gradient Boosting	Accuracy	82.41%
[36]	Classification	Logistic Regression	F1 Score	0.59705
		Decision Trees	F1 Score	0.59273
		Neural Network	F1 Score	0.67248
		Random Forest	F1 Score	0.66797
		Gradient Boosting	F1 Score	0.64848
[37]	Regression	Multiple Linear Regression	R^2 (Pre-DTAP)	0.63
			Std. Error (Pre)	3.12
			R^2 (Post-DTAP)	0.50
			Std. Error (Post)	5.08
[38]	Regression	XGBoost	RMSE	2.03
			R^2	0.89
[39]	Classification	Naive Bayes	Accuracy	Not Specified
		Random Forest	Accuracy	Not Specified
		Support Vector Machine	Accuracy	Not Specified
[40]	Neural Network	GPT-3	MAE	2.37 days
			MAE	2 days
			MAE	1.88 days

6. Discussion

This section presents a comprehensive comparison of various algorithms and metrics employed in different analyzed studies to evaluate their effectiveness and efficiency for medical data analysis [41]. Performance analysis is a crucial aspect of assessing the suitability of different methods for specific tasks, as it helps identify the strengths and weaknesses of each approach.

- Regression Models:
  - Study [28] demonstrated the superior performance of Ridge Regression (MAE = 0.82131) compared to other regression techniques, including logistic regression, Lasso, and ElasticNet;
  - In [30], random forest achieved the highest R^2 score of 0.7780, followed by gradient boosting and XGBoost;
  - Multiple linear regression models in [37] showed a higher R^2 (0.63) before the implementation of the diagnostic–therapeutic-assistance pathway compared to after (0.50).
- Neural Networks:
  - Neural network models in [29] achieved high accuracy rates (94.74%) after optimization through grid search and random search;
  - Study [40] utilized the GPT-3 language model for hospital LoS predictions, yielding an MAE of 2.37 days and a median absolute error of 2 days.
- Decision Trees and Ensemble Methods:
  - The decision tree model in [33] obtained an accuracy value above 80% when predicting patient admissions;
  - Gradient boosting in [35] outperformed other algorithms, including an accuracy value of 82.41%, highlighting its effectiveness when handling complex medical data.
- Survival Analysis:
  - The survival analysis methods employed in [32] achieved a concordance index of approximately 0.7, indicating moderate predictive power using patient information.
- Comparation of Advanced Algorithms:
  - The XGBoost algorithm in [38] demonstrated high predictive accuracy for total hip arthroplasty, with an RMSE of 2.03 and an R^2 of 0.89;
  - Study [39] found the support vector machine to be the most accurate model for predicting patient admissions in emergency departments, although specific accuracy metrics were not provided.

Based on the summarized results in Table 4, we can analyze and rank the algorithms to determine the best-performing methods for predicting the LoS in hospitals.

The XGBoost algorithm stands out as the best-performing method for hospital LoS prediction, providing the highest R^2 Score of 0.89. The neural network also demonstrates high accuracy and robustness across multiple studies, making it highly effective for hospital LoS predictions when optimized. Tree-based models, like the gradient boosting algorithm and random forest, also offer strong predictive capabilities, emphasizing their importance in healthcare analytics. These findings suggest that advanced ML algorithms, particularly XGBoost and neural networks, are highly suitable for predicting hospital LoS, potentially leading to better resource allocation and improved patient care.

Table 4. Best-performing algorithms.

Study	Best Algorithm	Metric	Values
[28]	Regression	Ridge Regression	0.82131
[29]	Grid Search Optimization and Random Search Optimization	Accuracy	94.74%
[30]	Random Forest	R^2 Score	0.7780
[32]	Survival Analysis (Various models)	Concordance Index	0.7
[33]	Decision Tree	Accuracy	Above 80%
[34]	Random Forest Regressor	Accuracy	59.78% and 36.57%
[35]	Gradient Boosting	Accuracy	82.41%
[36]	Neural Network	F1 Score	0.67248
[37]	Multiple Linear Regression (Pre-DTAP)	R^2 Score	0.63
[38]	XGBoost	R^2 Score	0.89
[39]	Support Vector Machine	Accuracy	Not Specified
[40]	GPT-3	MAE	2.37 days

6.1. Critical Analysis

While performance research emphasizes robust candidates, such as neural networks and XGBoost, an in-depth analysis has revealed context-dependent challenges and limitations. Neural networks excel in accurate predictions, as seen in [29], but require significant data preprocessing and parameter tuning to achieve an optimal performance. The heavy reliance on hyperparameter optimization (e.g., grid search in [29]) means that the performance can vary significantly between systems and datasets if tuning is not performed properly.

Similarly, XGBoost showed robust results (R^2 = 0.89 in [38]), but it is also computationally demanding and prone to overfitting on small datasets, making them valuable in healthcare contexts, where significant results are rare but can be surprising. Differences in study design and data collection characteristics between studies also influence the reported effectiveness of the algorithms. For example, the neural network underperformed in [34], showing a worse performance compared to random forest, possibly due to small feature sets. In contrast, the neural network in [29] achieved higher accuracy, with more features and broader tuning. This discrepancy highlights the importance of careful selection and validation to ensure that performance metrics reliably reflect the capabilities of the algorithm.

The complexity of neural network and ensemble models may make them less practical for immediate use in clinical settings. Although neural networks provide high accuracy (as previously mentioned in [29]), their "black-box" nature may challenge interpretations, which is a key factor in healthcare decision making. Practical implementations require improvements in interpretability or alternative methods, especially considering that healthcare environments require confidentiality and security. In addition, advanced methods, such as neural networks and XGBoost, have high computational requirements, which may not be feasible in some healthcare systems due to resource limitations.

Considering the flexibility and consistently high performance of the neural network algorithm in many of the selected studies, it appears to be a promising alternative for hospital LoS predictions. Its flexibility to handle different features allows it to perform well on different datasets, as supported by references [29,36]. However, it is critical to acknowledge that this success depends on extensive preprocessing and hyperparameter tuning, which may limit its usefulness in settings lacking technical expertise or computational resources.

Based on critical comparisons and analyses, neural networks exhibit strong adaptive capabilities and predictive capabilities for hospital LoS predictions, but successful applications depend on dataset quality and context-specific needs. They shows great promise

for improving patient care and predicting hospital LoSs, especially if the features are appropriately selected and the dataset is well preprocessed.

6.2. Answers to the Research Questions

Based on the RQs posed, after this comparison, they can be answered.

- RQ1: What are the key factors that significantly influence hospital LoS?
  - Patient Demographics: Age, gender, and other demographic variables;
  - Clinical Features: Diagnosis and severity of illness;
  - Operational Factors: Admission status and pre-operative hospital LoS;
  - Data Preprocessing Techniques: Feature selection, normalization, and handling of missing data.
- RQ2: What types of ML algorithms have been employed for hospital LoS prediction?
  - Regression Models: Linear regression, ridge regression, Lasso regression, ElasticNet regression, and multiple linear regression;
  - Neural Network: Various configurations of neural networks optimized through techniques like grid search and random search;
  - Decision Trees and Ensemble Methods: Decision trees, random forest, and gradient boost (e.g., XGBoost);
  - Bayesian Methods and Advanced ML Algorithms: Bayesian methods, support vector machine, and advanced ensemble techniques;
  - Text-Based Models: GPT-3 and other neural network language models.
- RQ3: How do these algorithms compare in terms of their performance metrics to predict hospital LoS?
  - ElasticNet Regression: Shows a superior performance among regression models, with a lower mean absolute error (MAE) compared to simple linear regression;
  - Neural Network: Achieves high accuracy rates (e.g., 94.74% in one study), but its performance is highly dependent on extensive preprocessing and parameter optimization;
  - Ensemble Methods: XGBoost outperforms traditional methods when handling complex relationships within large datasets, demonstrating lower prediction errors;
  - Bayesian Methods and Support Vector Machine: Provide high accuracy for non-disease-specific predictions, with the support vector machine often outperforming other models, like naive Bayes and random forest;
  - Text-Based Models (GPT-3): Show potential, but may not yet surpass expert human predictions in accuracy.
- RQ4: What are the potential benefits and limitations of using ML models for hospital LoS predictions in healthcare settings?
  - Benefits: Improved accuracy, operational efficiency, and data-driven decisions;
  - Limitations: Complexity, interpretability, data quality, and availability.

Our review of the literature revealed that the neural network algorithm is one of the most popular, with a high degree of adaptability and applicability to a wide variety of predictions. As a result, the analysis determined that the neural network algorithm is the best-performing algorithm in this study on the datasets tested. However, the selection of the optimal algorithm for hospital LoS predictions may depend on the specific characteristics of the dataset and the intended application context.

It is worth noting that the papers analyzed also described the use of other algorithms (as shown in Tables 3 and 4), but none of them stood out as much or achieved such positive results as the neural network algorithm.

7. Conclusions

In short, this article aims to provide greater knowledge about the prediction of hospital admissions and hospital LoSa using ML algorithms, having carried out, as a result, an analysis of articles related to the topic. Some ethical questions were raised, particularly related to the privacy of patient data and, therefore, it is important that these issues are properly considered before implementing any prediction system. Following the methodology and the subsequent comparative study of the selected papers, a critical analysis of the different algorithms was performed. It was concluded that the machine learning algorithm best suited for predicting hospital admissions and hospital LoSs is the neural network, which got the edge over other algorithms because it was the best-performing algorithm on the datasets being tested. Overall, the selection of the optimal algorithm for hospital LoS prediction depends on the specific characteristics of the dataset and the intended application context. Stakeholders should consider a judicious balance between accuracy, interpretability, and generalizability to achieve the desired outcomes for patient care delivery. Challenges associated with implementing hospital admission and hospital LoS prediction systems include the availability of quality data, the consideration of ethical issues, as well as data protection.

We also concluded that ethical considerations and potential challenges associated with implementing ML algorithms for hospital admission and hospital LoS predictions in practice are important, such as:

- Bias and fairness. Ensure that ML models do not perpetuate biases present in the training data, leading to the unfair treatment of certain patient groups;
- Transparency and accountability. The need for transparency in how models make predictions and accountability for decisions made based on ML predictions;
- Patient privacy in protecting patient data and ensuring compliance with the aforementioned regulations.

As future work, we intend to apply these methods and test the fittest models in a real hospital scenario. We also intend to apply these methods to different datasets.

Author Contributions: Conceptualization, F.B.C. and A.R.B.; methodology, F.B.C., A.R.B. and J.B.; validation, F.B.C., A.R.B. and J.B.; formal analysis, F.B.C. and A.R.B.; investigation, G.A.; resources, F.B.C. and A.R.B.; data curation, F.B.C. and A.R.B.; writing—original draft preparation, G.A.; writing—review and editing, F.B.C., A.R.B. and J.B.; visualization, G.A., F.B.C., A.R.B. and J.B.; supervision, F.B.C. and A.R.B.; project administration, F.B.C., A.R.B. and J.B.; funding acquisition, J.B. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Not applicable.

Conflicts of Interest: The authors declare no conflicts of interest.

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